Amazon API Quick Start Guide

GCC Database Solution

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Overview

This guide helps Amazon integrate with the GCC (Green Circle Certified) Partner API to retrieve certificate and product identifier data.

Integration Model: Amazon PULLS data from GCC using API keys.

1. Getting Your API Key

Contact Green Circle Certified to receive your partner API key. Keys follow this format:

```
gcc_partner_[32-character-hash]
```

Example: `gcc_partner_b451fb3b31c645e9dd900c3c7b2f6250fbb7480bbb4510d602efc5e52102fc4d`

> **Important:** Keep your API key secure. Do not share it in public repositories or client-side code.

2. Authentication

All API requests require the `X-API-Key` header:

```
curl -X GET "https://[production-domain]/api/v1/certificates" \
  -H "X-API-Key: YOUR_API_KEY" \
  -H "Accept: application/json"
```

Response Codes

Code	Meaning
200	Success
401	Invalid or missing API key
403	Endpoint not permitted for your key
404	Resource not found
429	Rate limit exceeded (200 requests/hour)

3. Data Access Scope

Your API key determines what data you can access:

Association Type	Data Access
Partner = Amazon	Certificates where Partner field = "Amazon"
Company associations	Certificates for those specific companies
Both Partner + Companies	Certificates matching EITHER (OR logic)
No associations	No data access (empty results)

> **Note:** You will only see certificates and product identifiers associated with your partner account. This is automatic - no additional filtering is needed.

4. Key Endpoints

Base URL

```
Production: https://[production-domain]/api/v1
```

Available Endpoints

Endpoint	Description
`GET /certificates`	List all accessible certificates
`GET /certificates?includeProductIds=true`	List with ASINs/UPCs
`GET /certificates/gs1`	**GS1 JSON-LD format (RECOMMENDED for Amazon)**
`GET /certificates/gs1/{number}`	Single certificate in GS1 format
`GET /certificates/{id}`	Get certificate by database ID

Endpoint	Description
`GET /certificates/number/{number}`	Get certificate by number (e.g., 24-0366)
`GET /certificates/search/productid`	Search by ASIN/UPC
`GET /certificates/{number}/download`	Download certificate PDF

5. Common Use Cases

5.1 GS1 JSON-LD Format (RECOMMENDED)

> **This is the primary endpoint for Amazon Climate Pledge Friendly integration.**

The GS1 endpoint returns data in the GS1 Web Vocabulary JSON-LD format, which matches Amazon's certification data requirements.

```
curl -X GET "https://[domain]/api/v1/certificates/gs1?pageSize=100" \
  -H "X-API-Key: YOUR_API_KEY" \
  -H "Accept: application/json"
```

Response:

```
{
  "success": true,
  "data": [
    {
      "@context": "https://gs1us-technology.github.io/certification-data-model/cert_context.json",
      "$schema": "https://gs1us-technology.github.io/certification-data-model/cert_schema.json",
      "id": "https://greencirclecertified.com/api/v1/certificates/gs1/24-0366",
      "type": "CertificationReport",
      "issueDate": "2024-01-15",
      "issuer": "Green Circle Certified",
      "products": [
        {
          "id": "https://id.gs1.org/01/00012345678901",
          "type": "Product",
          "gtin": "00012345678901",
          "additionalTradeItemId": [
            { "type": "ASIN", "value": "B08XQRT456" }
          ],
          "name": "Recycled Paper Towels",
          "description": "100% recycled paper towels",
          "certificationInfo": {
            "id": "https://greencirclecertified.com/api/v1/certificates/gs1/24-0366/certification",
            "type": "CertificationDetails",
            "agency": "Green Circle Certified",
            "agencyURL": "https://www.greencirclecertified.com",
            "program": "UL 2809",
            "certificateId": "24-0366",
            "standard": "UL 2809",
            "status": "ACTIVE",
            "value": "95.5% recycled content",
            "startDate": "2024-01-15",
            "endDate": "2025-01-15",
            "targetCountries": ["United States"]
          }
        }
      ]
    }
  ],
  "pagination": {
    "totalCount": 150,
    "pageNumber": 1,
    "pageSize": 100
  }
}
```


Key GS1 Fields:

- `@context` , `$schema` : JSON-LD metadata for schema validation
- `gtin` : Global Trade Item Number (14 digits, padded with leading zeros)
- `additionalTradeItemId` : Array of ASIN, SKU, MPN, ISBN identifiers
- `certificationInfo.status` : ACTIVE or INACTIVE
- `certificationInfo.value` : Certification results/level (e.g., recycled content percentage)

5.2 List All Certificates with Product Identifiers

Alternative endpoint with flat JSON structure.

```
curl -X GET "https://[domain]/api/v1/certificates?includeProductIds=true&pageSize=100" \
  -H "X-API-Key: YOUR_API_KEY" \
  -H "Accept: application/json"
```

Response:

```
{
  "success": true,
  "data": [
    {
      "id": 123,
      "certificateNumber": "24-0366",
      "status": "Active",
      "issueDate": "2024-01-15T00:00:00Z",
      "expirationDate": "2025-01-15T00:00:00Z",
      "company": "EcoProduct Co",
      "companyId": 45,
      "productName": "Recycled Paper Towels",
      "productId": 78,
      "certificationType": "UL 2809",
      "certificationTypeId": 3,
      "certificationTypeSubcategory": "Level 2",
      "certificationTypeSubcategoryId": 12,
      "issuer": "Green Circle Certified",
      "certificationResults": "95.5% recycled content verified",
      "productIds": [
        {
          "type": "ASIN",
          "value": "B08XQRT456",
          "status": "Active",
          "partner": "Amazon",
          "productName": "Recycled Paper Towels - 12 Pack",
          "productSku": "RPT-12PK-ECO",
          "description": "12-pack of 100% recycled paper towels"
        },
        {
          "type": "UPC",
          "value": "012345678901",
          "status": "Active",
          "partner": null,
          "productName": "Recycled Paper Towels",
          "productSku": "RPT-SINGLE",
          "description": "Single roll recycled paper towels"
        }
      ],
      "pdfDownloadUrl": "/api/v1/certificates/24-0366/download"
    }
  ],
  "pagination": {
    "totalCount": 150,
    "pageNumber": 1,
    "pageSize": 100,
    "totalPages": 2,
    "hasNext": true,
    "hasPrevious": false
  }
}
```


5.3 Search by ASIN

Find certificates containing a specific ASIN:

```
curl -X GET "https://[domain]/api/v1/certificates/search/productid?productId=B08XQRT456&type=ASIN" \
  -H "X-API-Key: YOUR_API_KEY" \
  -H "Accept: application/json"
```

Response:


```
{
  "success": true,
  "data": [
    {
      "id": 123,
      "certificateNumber": "24-0366",
      "productName": "Recycled Paper Towels",
      "productType": "Paper Products",
      "company": "EcoProduct Co",
      "companyId": 45,
      "partner": "Amazon",
      "certificationResults": "95.5% recycled content verified",
      "status": "Active",
      "certificationType": "UL 2809",
      "issueDate": "2024-01-15T00:00:00Z",
      "expirationDate": "2025-01-15T00:00:00Z",
      "pdfDownloadUrl": "/api/v1/certificates/24-0366/download",
      "productIds": [
        {
          "type": "ASIN",
          "value": "B08XQRT456",
          "status": "Active",
          "partner": "Amazon",
          "productName": "Recycled Paper Towels - 12 Pack",
          "productSku": "RPT-12PK-ECO",
          "description": "12-pack product",
          "isMatchedId": true
        }
      ]
    }
  ],
  "message": "Found 1 certificate(s) with 1 total product IDs for search: B08XQRT456"
}
```


> **Note:** The `isMatchedId: true` flag indicates which product identifier matched your search.

5.3 Get Certificate by Number

```
curl -X GET "https://[domain]/api/v1/certificates/number/24-0366" \
  -H "X-API-Key: YOUR_API_KEY" \
  -H "Accept: application/json"
```

5.4 Download Certificate PDF

```
curl -X GET "https://[domain]/api/v1/certificates/24-0366/download" \
  -H "X-API-Key: YOUR_API_KEY" \
  -o certificate.pdf
```

6. Field Mapping Reference

Certificate Fields

Amazon Requirement	GCC API Field	Notes
Product ASIN	`productIds[].value` (where type="ASIN")	Use `includeProductIds=true`
Brand name	`company`	Company/brand name
Product name	`productName`	Product name
Product type	`productType`	Product type (in search results)
Certificate ID	`certificateNumber`	Format: YY-NNNN
Issue date	`issueDate`	ISO 8601 format
Expiration date	`expirationDate`	ISO 8601 format
Certification type	`certificationType`	e.g., "UL 2809"
Standard version	`certificationTypeSubcategory`	May be null
Issuer	`issuer`	Always "Green Circle Certified"
Certification Results	`certificationResults`	Threshold/score data for compliance
Partner	`partner`	"Amazon" for filtered results
Status	`status`	Active, Expired, etc.
PDF download	`pdfDownloadUrl`	Relative URL path

Product Identifier Fields

Each item in `productIds[]` contains:

Field	Type	Description
type	string	ASIN, UPC, GTIN, EAN, ISBN, SKU, MPN
value	string	The identifier value
status	string	Active, Inactive, Pending, Expired
partner	string	Amazon, Novi, None, Other
productName	string	Product name for this specific ID
productSku	string	SKU for this identifier
description	string	Optional notes
isMatchedId	boolean	True if this ID matched search (search only)

7. Pagination

Use these query parameters to paginate results:

Parameter	Default	Max	Description
`pageNumber`	1	-	Page to retrieve
`pageSize`	20	100	Items per page

Example: Get page 2 with 50 items:

```
curl "https://[domain]/api/v1/certificates?includeProductIds=true&pageNumber=2&pageSize=50" \
  -H "X-API-Key: YOUR_API_KEY"
```

Pagination Response Fields:

- `totalCount` - Total number of certificates
- `pageNumber` - Current page
- `pageSize` - Items per page
- `totalPages` - Total pages available
- `hasNext` - More pages available
- `hasPrevious` - Previous pages exist

8. Rate Limits

Metric	Limit
Requests per hour	200
Requests per minute	~3

When rate limited, you'll receive HTTP 429. Implement exponential backoff in your integration.

9. Testing Checklist

Use this checklist to validate your integration:

Test	Status
Authenticate with API key (expect 200)	☐
Test invalid API key (expect 401)	☐
List certificates with `includeProductIds=true`	☐
Verify ASIN values appear in `productIds[]`	☐
Search by a known ASIN	☐
Verify `certificationResults` field is present	☐
Download a certificate PDF	☐
Test pagination (page 1, page 2)	☐
Verify date formats are ISO 8601	☐

10. Sample Test Script

PowerShell

```
$apiKey = "YOUR_API_KEY"
$baseUrl = "https://[production-domain]"

$headers = @{
    "X-API-Key" = $apiKey
    "Accept" = "application/json"
}

# Test authentication
$response = Invoke-RestMethod -Uri "$baseUrl/api/v1/certificates?pageSize=1" -Headers $headers
Write-Host "Total certificates: $($response.pagination.totalCount)"

# List with product IDs
$response = Invoke-RestMethod -Uri "$baseUrl/api/v1/certificates?includeProductIds=true&pageSize=5" -H
foreach ($cert in $response.data) {
    Write-Host "Certificate: $($cert.certificateNumber) - $($cert.productName)"
    foreach ($pid in $cert.productIds) {
        if ($pid.type -eq "ASIN") {
            Write-Host "  ASIN: $($pid.value)"
        }
    }
}
```

Bash/cURL

```
API_KEY="YOUR_API_KEY"
BASE_URL="https://[production-domain]"

# Test authentication
curl -s -H "X-API-Key: $API_KEY" "$BASE_URL/api/v1/certificates?pageSize=1" | jq '.pagination.totalCou

# List with product IDs
curl -s -H "X-API-Key: $API_KEY" \
  "$BASE_URL/api/v1/certificates?includeProductIds=true&pageSize=5" | \
  jq '.data[] | {cert: .certificateNumber, asins: [.productIds[] | select(.type=="ASIN") | .value]}'

# Search by ASIN
curl -s -H "X-API-Key: $API_KEY" \
  "$BASE_URL/api/v1/certificates/search/productid?productId=B08XQRT456&type=ASIN" | jq '.'
```

11. Error Handling

Error Response Format

```
{
  "success": false,
  "errors": [
    {
      "code": "ERROR_CODE",
      "message": "Human-readable error message"
    }
  ]
}
```

Common Error Codes

Code	Description	Resolution
`UNAUTHORIZED`	Invalid or missing API key	Check X-API-Key header
`FORBIDDEN`	Endpoint not permitted	Contact GCC for permissions
`RESOURCE_NOT_FOUND`	Certificate/resource not found	Verify ID or number
`VALIDATION_ERROR`	Invalid request parameters	Check query parameters
`RATE_LIMIT_EXCEEDED`	Too many requests	Wait and retry with backoff

12. Support

For API issues or questions:

- **Technical Support:** [Contact GCC Technical Team]
- **API Key Issues:** [Contact GCC Admin]
- **Documentation:** See full API reference at `/swagger` (development only)

Appendix: Quick Reference

Minimum Required Headers

```
X-API-Key: YOUR_API_KEY
Accept: application/json
```

Most Common Request

```
GET /api/v1/certificates?includeProductIds=true&pageSize=100
```

Certificate Status Values

- `Draft` - Not yet active
- `Active` - Currently valid
- `Expired` - Past expiration date

- `Suspended` - Temporarily inactive
- `Revoked` - Permanently invalidated

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